

Evidence for Dose De-escalation in Brachytherapy for Choroidal Melanoma



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Keywords

- Uveal melanoma • Choroidal melanoma • Brachytherapy • Dose reduction
- Brachytherapy complications

Key points

- Brachytherapy has been shown to be a safe and effective globe-conserving therapy for uveal melanoma. At the current standard dose of 85 Gy, modeled after the Collaborative Ocular Melanoma Study (COMS) trials, brachytherapy is associated with high rates of radiation-related complications.
- There is evidence that a prescription dose of less than 85 Gy for brachytherapy of uveal melanomas would maintain equivalent rates of local tumor control.
- Reduction of radiation doses to critical ocular structures may result in a clinically significant improvement in rates of complications and rates of visual decline.
- Prospective dose de-escalation trials using ¹⁰⁶Ru brachytherapy, proton beam, and gamma knife radiosurgery have demonstrated favorable results.
- A randomized prospective trial is necessary to determine a reduced prescription dose for ¹²⁵I brachytherapy that would maintain acceptable local tumor control while minimizing dose-dependent radiation complications.

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